

LMAP NOTES

A collaboration project of ICSM
Paediatrics Society, ICSM Psychiatry
Society and ICSM Oncology Society



Created by Kavyesh Vivek, Hana Isphani, Piyush Koorapaty and Sara Alikhan

Measuring & describing a disease

Epidemiology:

- The study of the **distribution & determinants** of health-related states/events in **specified populations** & **application** of this study to the control of health problems
- How** often diseases occur in different groups of people and **why**?
- Endemic** – diseases that reside **within** a population
- Epidemic** – diseases that **befall** a population

Types of prevention:

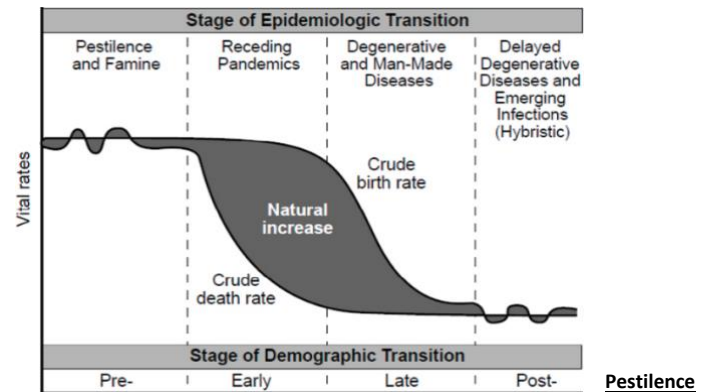
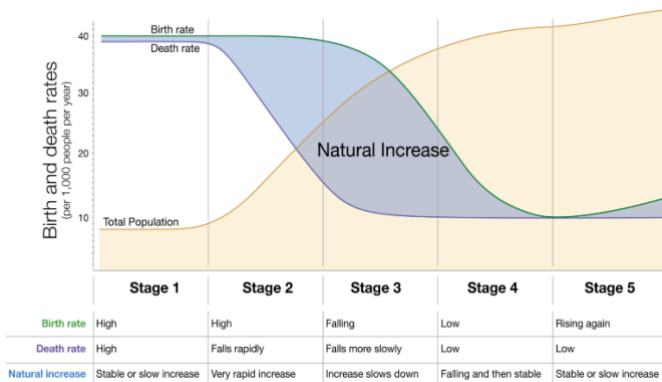
- Primary (Before onset)**: Prevention through the **control of exposure to risk factors** (↓salt→↓HTN risk)
- Secondary (Slows progression)**: Application of available measures to **detect early departures from health** & to **introduce appropriate treatment & interventions** (Antihypertensives↔HTN)
- Tertiary (Return to function after insult)**: Application of measures to **reduce/eliminate long-term impairments & disabilities**, minimising suffering caused by existing departures from good health & to promote the **patient's adjustments to their condition** (rehabilitation↔stroke↔pre-morbid activities)

- Epidemiology requires**: Curiosity, Problem solving, Numerical ability, Critical thinking, Communication and Creativity

- Questions**: What, Why, When, How, Where, and Who & Time, Person, and Place

Demographic transition model:

Epidemiological transition model:



and famine (Pre-Industrial Revolution in UK (up to 1800)): **Urbanisation**. Food supply constraints. **High birth rate & mortality**. Low life expectancy at birth.

Receding pandemics (1800s - c.1950 in UK): **Agricultural development** improves nutrition. **Water, sanitation, hygiene & vaccination** emerges. **High birth rate & reducing mortality**. Increasing life expectancy.

Degenerative & man-made diseases (c.1950 - 2010s in UK): **Lifestyle factors & NCDs** (cancer and CVD) **predominate**. Environmental & global determinants drive **obesity** and other **risk factors**. Technology reduces need for physical labour. Addiction, violence & other issues emerge.

Delayed degenerative diseases & emerging infections (2010s onwards): **Health technology defers morbidity**, albeit at increasing financial cost. Emerging **zoonotic disease** presents new threats. **Inequalities** within & between countries come to the fore.

Exposures and outcomes:

- Exposure (independent variable)**: variable we are trying to **associate** with a **change** in health status e.g. drug exposure
- Outcome (dependent variable)**: variable we are **measuring** e.g. mortality at five-years
- Can be a drug, behaviour, or demographic characteristic.

Population pyramids:

Like Yemen - rapidly growing



Like the UK/NZ – immigration



Like Spain



DALYs (Disability Adjusted Life Years): measure of **disease burden** that combines years of life lost from **ill-health, disability, or premature death**

Three groups of conditions: **Communicable disease, non-communicable & injuries**

The burden of disease:

- Mortality and morbidity different**. Don't be driven purely by what kills. QOL matters e.g. opioid epidemic.
- Disease distributed differently between **sex & age groups**.
- No health without **mental health**. Self-harm, substance misuse, anxiety & affective disorders are important causes of both **morbidity & mortality**. Mental health not merely 'nice-to-have'.
- Population health is about understanding both the **quantum & distribution of health** across a population.
- Inequalities** - Most disadvantaged demonstrate higher levels of disease & have fewer tools to mitigate disease impact.

Measures of frequency for discrete variables:

Odds = No. of people with disease / No. of people who don't have disease

Prevalence = No. of people with the disease / Total no. of individuals in the population

- Proportion** of individuals in a population who have the disease/attribute of interest **at a specific timepoint**
- Dimensionless**, given as a percentage/number from 0-1
- Reflects **occurrence & duration** of disease but **no** information on **new cases**
- Prevalence & odds** can describe the health of a population & monitor trends over time. Enable planning of health services & allocation of healthcare resources. Less helpful in diseases of short duration & causal inference.

Cumulative incidence = No. of new cases during period of interest / No. of disease-free individuals at start of time period

- Dimensionless. Proportion of the population with a new event **during a given time period**.
- Alternative/interchangeable terms = **Risk/Incidence Proportion**
- Requires **follow up**, must be the same for all participants, during follow up no new participants can join, inaccuracy occurs if participants are lost to follow up (e.g. death from an unrelated disease)

Incidence rate = No. of new cases during the follow-up period / total person-time by disease-free individuals

- **Person-time:** Time participants spend in study. Stops when diagnosis of interest is made, participant dies or is lost to follow up. Range of units e.g. person-years, person-days, etc.
- Always expressed by unit of person-time. Rates can only be expressed as **new cases per unit of person-time**.
- Accounts for time of follow up and for time when new event occurred.
- Suitable when participants enter and leave study at different times, participants are lost to follow up, there are competing risks.

Types of research:

Levels of evidence: Refer to Figure 1 (N.B. Top layer should say **systematic reviews AND meta-analysis**)

Research methods: Qualitative & quantitative

Study design: Observational & interventional

Epidemiological approaches: Descriptive & analytical

Qualitative research:

- Explores underlying ideas & themes to inform research questions & possible **hypotheses**
- Expresses its findings & outputs **in words**
- Relies on **smaller numbers** of participants, but goes in substantial detail



Figure 1

What numbers and measures do we use?

- Measures of **frequency & association**
- Comparisons & adjusting for differences

Where do these measures come from?

- **Descriptive epidemiology**
- **Observational & interventional** study design

- **Systematic reviews & meta-analysis**

How do we interpret epidemiological findings?

- **Association, causation, validity, & bias**
- **Confounding & effect modification**

Direct and Indirect standardisation:

Direct - similar incidence e.g. 120 strokes/100,000 people/year

Indirect - ratio out of 100 or 1.0

Direct standardisation:

- Adjustment that allows us to compare like-for-like between populations as differences may be due to differences in demographics i.e. **sex & age**.
- Steps for age specific (sex-specific requires more steps):
 - Calculate age specific incidence (rate)
 - Rate $\times$ Standard population (for each age category) = expected count for each age category
 - Total overall standard population & expected count
 - Total expected count / total standard population = age-standardised incidence
- Questions to ask after results:
 - Could it be low numbers and therefore chance?
 - Are people healthier?
 - Is there something missing in terms of prevention?
 - Are there inequalities?
 - How do these two areas compare to anywhere else?

AGE GROUP	TOWN A			TOWN B		
	Rate*	Standard Popn	Expected	Rate*	Standard Popn	Expected
0-15 years	25.0	15000	3.8	9.1	15000	1.4
16-64 years	72.0	48000	34.6	53.8	48000	25.8
65-79 years	266.7	14000	37.2	373.3	14000	52.3
80 years +	400.0	5000	20.0	550.0	5000	27.5
TOTAL*		82000	95.6		82000	106.9
	Age-standardised incidence*			Age-standardised incidence*		
TOTAL*		E/Popn=	116.6		E/Popn	130.4

Indirect standardisation:

- Use if no age/sex specific data.
- Crude rate/ raw / unadjusted
- Unusual results caused by:
 - **Unwarranted variation** (a hospital is dangerous)
 - **Explained variation** (a hospital has a higher risk blend of procedures)
 - **Statistical artefact** (a hospital is better at recording deaths than other hospitals)
- **Standardised Mortality Ratio (SMR)** = observed count/ expected count. <0.75 = marker of healthy life expectancy.
- **Standardised incidence ratio (SIR)** = observed incidence/ expected incidence
- **Summary Hospital-Level Mortality Indicator (SHMI)**: Used to identify hospitals that report higher than expected mortality (ranging from 0.6-1.2). At level of organisation e.g. ICHT hospital groups not CX individually.
- Indirect standardisation is useful when we have only **high-level data** about outcomes, but we **can't make a direct comparison**.

Incidence vs Incidence rate: Incidence rate is divided or averaged across a unit of time.

Incidence rate vs Cumulative incidence: Incidence rate is a rate and cumulative incidence is a proportion.

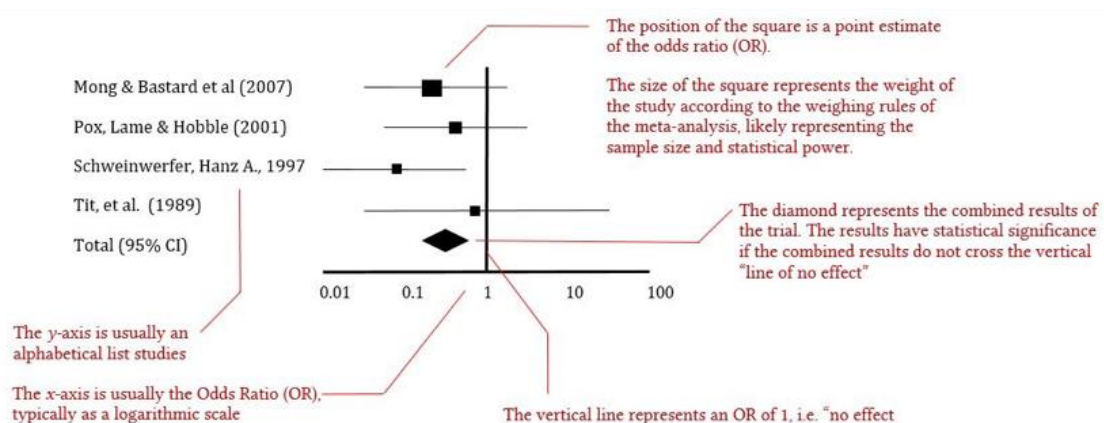
Positive findings:

- Could it be chance? Could it be bias (including confounding)?
- If neither of the above, then it may be a true positive.
- **Check for:** Chance finding, bias, confounding, confidence intervals

Designing epidemiological studies

- **Statistic:** a fixed value, derived from a sample that estimates the value in the population (so estimates parameter)
- **Parameter:** a fixed, often unknown value, which describes an entire population
- **Point estimate:** an estimate of a parameter (a statistic)
- **Confidence interval:** the range of values in which there is a 95% confidence that the true value lies within these
- **Data linkage:** joining datasets (>2) together. Can find out more than each dataset separately. Technical issues & privacy concerns.
- **Null hypothesis (H_0):** no effect or relationship between variables
- **Alternative hypothesis (H_1):** states your research prediction of an effect or relationship
- **Randomised clinical trials:**
 - Gold standard of investigations. Compare effectiveness of 2 or more treatments with a control group (placebo) for comparison $\rightarrow$ **placebo-controlled trials**.
 - Can be testing **effectiveness and side effects**. Choice of controls is critical.
 - Participants are allocated to study groups by **randomisation**, but this does not fully protect against bias as it can be introduced by poor analysis of data.

- **Intention to treat** - patients are analysed based on their originally assigned groups, regardless of whether they actually followed the allocated intervention.
- **Randomisation process:**
 - All participants have equal chance of being assigned to each group
 - Key component of **avoiding/limiting selection bias**
 - Not determined by investigators, clinicians, or participants
- 1. **Generate allocation sequence:** simple randomisation (not suitable for large numbers; random number generator)
- 2. **Allocation concealment**
- 3. **Implement allocation:** sequentially numbered opaque envelopes (random, opened on enrolment by recruitment); central randomisation (randomisation services contacted on enrolment)
- **Blinding:**
 - Procedure where one or more parties are kept unaware of participants' allocations
 - To prevent conscious or unconscious bias in trial design and delivery
 - Reduces placebo effect and patient withdrawal but not always possible
- **Sources of bias:** Participant; Clinical staff administering treatment; Assessing physician; and People interpreting result
- **Single blinding:** only one party is blinded
- **Double blinding:** participants and staff are both blinded, particularly useful for preventing performance and detection bias
- **Performance bias:** systemic differences between groups in care that is provided, or in exposure to factors other than the exposure of interest (e.g. clinicians focus more on group receiving drug)
- **Detection bias:** **systematic differences between groups in how outcomes are determined** (e.g. more frequent examinations of group receiving the drug under investigation)
- **Power and sample size**
 - **Sample size** indicates how many participants you need to recruit to achieve sufficient power & is affected by withdrawal. It is calculated before the trial starts.
 - **Power** is the ability of a study to detect effect/association if one exists (80% + (preferably 90%)), **High sample size = High power**
 - **Alpha** is how much you want to rule out chance causing a positive finding, equivalent of specifying **p-value** & connected with one/two tailored testing
 - **Sample size** affected by: **difference between groups** (high difference doesn't need such a big sample size); **power**; **alpha** (setting the **p-value**- how carefully chance is ruled out); decreasing alpha will increase sample size; and **loss to follow up**
- **Errors**
 - **Type I (false positive)**
 - **p-value** is the probability of a positive result being due to chance alone
 - **alpha** is the probability of making a **type I error**
 - statistical significance not always clinically significant; greater number of statistical tests performed increases type 1 error
 - **Type II (false negative)**
 - statistical **power** is important
 - going from 80% to 90% power halves the risk of **type II error**, but needs a much larger sample
 - **beta** is probability of making **type II error**; increase power reduces **type II error**; doubling **sample size** reduces **type II error**
- **Process for systematic review:**
 1. Research question
 2. Structured search (series of search terms which can be put into a search engine)
 3. Indices (e.g. Medline, must be clear about which is used)
 4. Registries (early stage where researchers register their research to avoid duplication or omission, prevents publication bias)
 5. Screening (**PRISMA diagram** shows how many articles found in initial search, how many were duplicates, how many were eligible)
 6. Reporting
 7. Writing
 8. Submitting, revising, publishing
 - Not everything that is known is necessarily in published literature: that which has been reviewed & published within academic journals that can be searched via online indices (**grey literature**)
- **Meta-analysis and forest plot**



- Quantitative, formal epidemiological study design used to systematically assess previous research studies to derive conclusions about the body of research
- At top of pyramid in terms of quality of information and level of evidence
- Combines quantitative findings from multiple studies into a pooled estimate presented as a **forest plot**
- **Components:** study details, intervention group, control group, forest plot, weighing (**fixed or random effects model**), 95% CI
 - Vertical line= **line of no effect**
 - Each square box= **point estimate** (RR or OR of individual studies)
 - Square sizes **proportional to number of participants/weights of each trial**
 - Whiskers show **level of confidence or uncertainty** around each estimate
 - Width of CI indicates precision of estimate; wider interval = less precision
 - If CI includes 1: not statically significant difference

- Higher number of patients and larger squares → whiskers shrink
- Position of squares either side of line of no effect indicates what is going on
- Final row aligned with total is a **diamond** → **pooled estimate** (combines power of each of studies and uses left and right angles to show **level of uncertainty**)
- **Diamond on Left side of line of no effect** = *Less episodes of outcome of interest in treatment group*
- **Diamond on Right side of line of no effect** = *More episodes of outcome of interest in treatment group*
- **Diamond touches line of no effect** = *Not statistically significant difference*
- **Diamond does not touch line of no effect** = *Difference between 2 groups is statistically significant*
- **Heterogeneity**
 - Meta-analysis requires that pooled studies are sufficiently similar
 - **Heterogeneity**: difference between studies that are included
 - 3 types of **heterogeneity**:
 - **Clinical**: difference in patient groups, selection criteria
 - **Methodological**: design, blinding, collecting/recording
 - **Statistical**: reporting differences
- **Publication bias**
 - Meta-analysis can be subject to publication bias - **studies with positive findings are more likely to be submitted for publication**
 - Managed via **funnel plot**
 - Assess likelihood of publication bias in sample; show balance of evidence between studies assuming an overall effect size: **more publications on one side of the line may suggest some studies may not have been reported**
- **Survival analysis**: Whenever a patient dies, their death is recorded at that time and a survival time is calculated
- **Kaplan Meier plot**: displays survival analysis, Shows proportion of participants alive at any particular point

Narrative review	Systematic review
• Complies published literature into a single article to enable more rapid understanding (aka literature scoping reviews, non-systematic reviews, critical review)	• Highly structured approach to searching, selecting, and summarising literature Underpin metanalysis
• Fast • More up to date • Useful if limited research or many fields involved • Bias - authors free to select work in line with their views • often no search specified	• Robust and reproducible • Structured method - includes evidence based on pre-specified inclusion criteria • Slow • Gets outdated quickly • Only as good as method used

Descriptive		Analytical	
• Describe the problem (often aggregate level) + can be used to inform analytic research • Provides measures of frequency		• Deploy and test hypotheses (often person-level) which allows to measure association and infer causation	
• Case report - often used for new diseases • Case series - multiple case report • Cross-sectional - describes prevalence of exposure/outcome across a pop. at a single point in time (no follow up + only prevalence, easy, cheap, and fast) • Longitudinal - cross sectional studies repeated over time with same participants and same questions; describes prevalence/incidence of an exposure/outcome over time (aggregated or person -level) • Ecological - compare groups rather than individuals (can be longitudinal/cross-sectional)		• Observational: case control studies, prospective cohort studies • Experimental: randomised controlled trials	
Primary data	Secondary data	Routinely collected	Non-routinely collected
• Collected by the researcher first-hand for a specific purpose • High cost • Time consuming	• Collected originally for a different purpose • Faster + cheaper • Have to make assumptions as data wasn't collected for the new purpose	• Mainstay of day-to-day epidemiology. • Large, readily available administrative databases e.g. census, electoral register, prescribing data, hospital episode statistics	• Primary data • Surveys and other bespoke data sets • Limited use • Expensive • Time consuming

Study design	What it is	Purpose/use	Advantages	Disadvantages
Case reports and series	- Report = short write up on findings on disease - Series = collection of case reports - Description of a constellation of signs & symptoms can be classed as new disease/syndrome	Often used to communicate new disease, new presentations, or new findings	More accessible end of publishing spectrum for medical students	
Cross sectional	- Typically describes prevalence of condition across population at a single point in time (e.g. a survey) - Assess outcome & exposure at same time - Prevalence measured may be an outcome, exposure, or both	- Can assess prevalence of disease or condition but not incidence rate or risk - No follow up involved	- Relatively cheap & easy to conduct - Provide info on multiple exposures & outcomes - Assess health needs of a population	- Lacks follow up so risk of temporal relationships cannot be easily determined, also no causality - May involve several weeks data collection, assume point estimate related to particular point in time, typically a month or a year - Unable to determine if exposure preceded outcome - Limit ability to measure true extent of disease as some may no longer have it or may have died
Longitudinal	- Describe prevalence or incidence of an exposure or outcome over time - May be made up of more than one cross-sectional analysis aggregated data - May look to follow same patients over time a person level data	Prevalence of disease amongst same patients over time, establish longitudinal trend		
Ecological	- Compare/measure groups rather than individuals - Can be descriptive or analytic in nature, cross sectional or longitudinal - Unit of observation is the group	- Can generate hypothesis about disease aetiology - Typically use secondary data sources that are already available	- Good for when variability of an exposure is limited - Good for when person level data isn't available	- Only get aggregate level data - Assuming associations between groups hold for individuals is called ecological fallacy or aggregation bias - Not always comparable between countries and time periods - Might also be unclear if exposure preceded the outcome
Case control studies	- Type of observational RETROSPECTIVE study - Comparing individuals with disease (CASES) to group with same characteristics but without condition (CONTROL) - Info on past exposure obtained for both groups - Frequency and intensity of exposure is compared 1. Clear eligibility – case definition 2. Select cases, should be representative of the condition 3. Select controls, should be representative of population at risk and from same study population as cases; assessment of exposures- should be measurable to a similar accuracy to exposure within cases 4. Assess exposures	- Can't calculate incidence based on exposure because you already know who has outcome - Estimating risk in control study: estimate likelihood of having exposure in those who have diseases relative to those who don't Group based on outcome first, then check for exposures ODDS RATIO	- Good for studying rare diseases - Relatively inexpensive to conduct - Quick to obtain data	- There can be bias associated with exposure assessment - Difficulty selecting control group - Limited to one outcome - Cannot provide info about temporal relationship between an exposure and disease - Self-reported behaviour usually more accurate in cases vs controls=> recall bias
Cohort studies	- Type of analytic PROSPECTIVE (usually) study; most robust observational study - Observe group of people over a period of time to look for outcomes (longitudinal study) 1) Select target population & assess exposures 2) Track over time with regular follow-up; going forward in time, exposure a outcome	Group based on exposure first, then check for outcome - Can identify cohort by location, occupation, disease, risk group, birth year - Assess for multiple exposures to control for those not of interest (potential source of confounding)	- Time sequence of events can be determined, useful for causality - Collection of info of several outcomes	- High cost - Large number of people need to be followed up over time - Not suitable for rare diseases - Ensuring people who start study stay until end

		- Exposures must be well defined and assessed by self-report, measurement, or records - Outcomes assessed by routine data surveillance or directly from participants but same method for exposed and unexposed groups RELATIVE RISK	and risk factors can be collected	
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Interpreting epidemiological studies

- | | | |
|---|--|---|
| <ul style="list-style-type: none"> Koch's Postulates Bradford Hill-Criteria Strength Consistency Specificity Temporality Biological gradient Plausibility Coherence Experiment Analogy | <ul style="list-style-type: none"> Internal validity External validity/generalisability Selection bias Berkson's bias Healthy worker effect Non-Response Bias Information Bias Interviewer bias Recall bias Response Bias Diagnostic Bias | <ul style="list-style-type: none"> Lake Wobegon Effect (Illusory Superiority) Hawthorne effects Confounding Confounding variable Effect modification Breslow-Day test Q test Interaction terms in regression models Synergism Antagonism Kaplan Meier Plot |
|---|--|---|
-
- Inferring Causation
 - Association does not equal causation
 - Bias
 - Bias** is any trend in the collection, analysis, interpretation, publication, or review of data that can lead to conclusions that are **systematically** different from the truth (not the same as random error as that can be corrected for by increasing sample size)
 - Bias** can harm validity
 - 3 types of **bias**
 - Selection Bias
 - Individuals' chance of being included in the study sample is related to both the exposure & the outcome
 - How to reduce Selection Bias:
 - Controls representative of target population
 - Minimise non-response
 - Information Bias
 - misclassification of exposure or disease status or both e.g. some heavy smokers classified as light smokers or lung cancer patients receiving wrong diagnosis
 - information bias typically arises from study variables not being properly defined or flaws in data collection
 - 2 specific types of information bias
 - Interviewer
 - Recall Bias
 - Confounding
 - Compare respondents with non-respondents

Internal Validity	External validity	Differential misclassification	Non-differential misclassification
Association truly exists amongst a specific group of patients	association in sample population can be applicable to other groups; generalisability	Results in a bias towards or away from the null	Results in bias towards the null hypothesis Due to exposure status is misclassified but equally so amongst cases and controls

Selection Bias	Berkson's Bias	Healthy Worker Effect	Information bias	Interviewer bias	Recall Bias
Individuals' chance of being included in the study sample is related to both the exposure & the outcome	A hospital-based case control study & controls are selected amongst the hospital's patients	Active workers more likely to be healthy compared to those who have retired or stopped working	Misclassification of exposure or disease status or both	Interviewer changes how they conduct interview based on disease category of participant – can be prevented by if interviewer doesn't know status of individual or interview protocols strictly defined	Patients recall different details of their history based on their disease status

Concept	Confounding	Effect Modification
Definition	A third variable which leads to bias in the estimate of association between outcome and exposure Has to be controlled for! Criteria for Confounding: 1) Associated with Exposure in Source Population 2) Associated with the Outcome in Absence of Exposure 3) Not a consequence of the exposure (not in causal path between exposure + outcome)	A third variable that has an impact that is so much it should not be dismissed as confounding Doesn't have to be controlled for! Synergism → potentiates effect of exposure Antagonism → diminishes effect of exposure
Methods we can identify it by	Knowledge of the subject matter - you could undertake an evidence review and see what other people have proposed or found – i.e. other studies indicate there are certain confounding variables Test by the three criteria (listed in box above) Stratification → pooled estimate very different to stratum specific estimates Compare crude & adjusted estimates → difference by 15% or more = indicates confounding N.B. Only one method required to identify confounding	Breslow-Day Test Q test Interaction terms in regression models
What to do if it is discovered?		Conduct stratified analysis & acknowledge in paper
Examples of Confounding	Alcohol & Lung Cancer – Tobacco consumption (confounding variable) which is associated with both alcohol consumption & lung cancer → the association is not actually between alcohol & lung cancer but rather tobacco and lung cancer	You conduct a RCT testing a new antibiotic against pneumonia. Patients split into 2 groups those with and without drug, they are 2 outcomes they either recover or die. You find there is an Odds Ratio of 1.5 -> i.e. those taking antibiotic 50% more likely to recover. But before you publish results a colleague stratifies the results finding that the OR is 1.1. for men and 1.9 for women. After careful consideration you determine bias is not playing a role -> Effect modifier = sex

Mnemonic	Bradford Hill Criteria	Explanation	Example/Additional Info
Clowns	Consistency	Across different settings	
Pursuing	Plausibility	Relationship shouldn't be implausible	
Epidemiology	Experiment	Evidence, not always ethically viable	Observational studies won't prove causation conclusively – so experimentation required but may not be ethically suitable
Are	Analogy	Analogous finding → make inference on the relationship	
Commonly	Coherence	Coherent with broader scientific understanding	
Behind	Biological gradient	Dose-response relationship	Linear/broadly linear relationships between exposure + outcome – e.g. more cigarettes congers greater risk!
The	Temporality	Exposure must precede an outcome, healthy worker effect	Cross-Sectional studies problematic for this reason → Longitudinal studies preferred
Silly	Strength	Stronger relationship more likely causation	Consistent findings across settings rule out errors or fallacies that might be an issue for one or 2 studies
Samples	Specificity	E.g. specific disease amongst specific workers	Most criticised, difficult in multi-factorial relations

Population health & inequalities

- Women's inequalities:**
 - Women are more likely to
 - Move into residential care
 - Provide informal, unpaid care for relatives and children
 - Be obese
 - Fat tissue is inflammatory, produces sex hormones (oestrogen), produces growth hormones (insulin, GF etc.)
 - Promotes rapid cell division → Increases cancer risk
 - Breast and endometrial cancers have closest link to obesity
 - Have dementia
 - Oestrogen protects brain cells but decreases post menopause
 - Be less physically active
 - Barriers = fear of judgement, lacking confidence, lacking time
 - Be misdiagnosed following MI – doctor bias
 - Be unrepresented equally in research
 - Have lower paid + lower skilled jobs – less likely to be financially secure
 - Women and men do different jobs; Men hold more of the most senior roles; Jobs done by women are undervalued
 - Women pay a 'motherly penalty'
- Tudor-Hart's inverse care law** states that more deprived areas tend to have poorer services (and fewer doctors).

Sleep, stress & dependence

- Physiology of sleep:**
 - Orexin – effects (higher levels in those who don't need much sleep)
 - Arousal state! (cerebral cortex activation + involves LC neurons)
 - Autonomic function
 - Food intake
- Effect of age on sleep:**
 - Sleep time decreases
 - Becomes more fragmented
 - Changes in type
 - From 2 months = circadian sleep (cycles every 24 hrs), infants = ultradian sleep (cycles within 24hrs)
 - Circadian rhythm delays in adolescents (stay up later and sleep in later)
 - Accompanied by increased light sensitivity (increased impact of screen time etc.)
- Sleep deprivation:**
 - Nocturnal enuresis can be result of sleep deprivation (rebound in REM sleep).
 - Changes in sleep architecture may be one way through which stress and anxiety also associate with bed-wetting.
 - Other indications = hyperactivity, behavioural problems at school and inability to focus, increases in interpersonal conflict, more impulsive behaviours, more labile mood.
 - In adolescence sleep deprivation also associated with more negative thinking and increased catastrophising
- Early intervention:**
 - Experience violence against women and girls
 - More likely to experience sexual/physical abuse, neglect, domestic violence
 - Men inequalities**
 - Shorter life expectancy** (avg. 5 years less)
 - Gap between life expectancies is greater in higher income countries
 - Suffer life-limiting disease sooner
 - Higher mortality rate
 - More accidental injury + interpersonal violence
 - Less likely to show health seeking behaviours
 - Put more stigma on mental health + less socially connected
 - More prone to **substance misuse, alcohol dependence, and homelessness**
 - Suicide and Mental health**
 - 3 in 4 suicides are men (women more likely to attempt)
 - Method of suicide is more likely to be violent
 - Presents signs of depression differently (more irritable)
 - Interventions**
 - Men are more likely to report lack of meaningful social connections
 - Workplace health initiatives
- Hormonal changes
- Glucose control
- Energy expenditure + metabolic rate
- Post adolescence – circadian system advances (genetics, neuropeptide expression, lifestyle)
- Slowed melatonin production in elderly (unless remained in good health)
- Ageing associated with yellowing of the lens and so may decrease amount of light that enters the system.
- Sleep disorders: insomnia and OSA most common
- In adolescence – sleep walking, night terrors, sleep paralysis, nocturnal enuresis

- Early intervention e.g. HENRY to reduce stress, build responsive relationships and strengthen life and coping skills can prevent the long term effects of ACEs and promote a healthy start to life.
- Giving a child the best start in life enables individuals to thrive and:
- Reduces health inequalities
- Prevents chronic disease
- Improves health and wellbeing
- Increases employment
- Increase tax revenue
- Reduces pressure on the health system

Relationships & finance

- **Impact of good quality relationships:**
 - Oxytocin released in response to social connection, sex, labour + childbirth
 - Promotes social bonding, prosocial behaviour, empathy, elevates mood, reduced stress
 - Live longer + healthier lifestyle
 - Suffer less from mental health issues + better recovery from mental/physical stressors
 - Better cardiovascular health + stronger immune system
 - Lower severity of perceived physical pain
 - Women more likely to have broader/more intimate friendships
 - More likely to share personal matters + feel more supported
 - Women more affected by -ve aspects of close relationships
- **Domestic abuse:**
 - Sexual/emotional/ psychological/financial
 - During pregnancy – increased risk of:
 - Miscarriage, infection, premature birth, postnatal injury/death
- **UCLA loneliness scale**
 - Types of connections:
 - **Intimate connections** - people who personally love and support you eg. family, friends, partners
 - **Regional connections** - people who you see on a regular basis and may share an interest with eg. work colleagues
 - **Collective connections** - people who share a common affiliation or group membership with you eg. people of your faith group

Behaviour change in lifestyle medicine

Brief interventions

- **MECC – Make Every Contact Count:**
 - **Smoking, alcohol, healthy eating, physical activity, healthy weight, mental health, and wellbeing**
 - **Benefits:**

Organisations	Staff	Individuals	National/Population	Community/Local
Meet obligations and core responsibilities towards local population health and wellbeing within NHS standard contract Meet responsibilities towards workforces e.g. training environment, staff awareness, staff skills & leadership, confidence, motivation, potentially improve staff health and wellbeing Can enhance existing initiatives e.g. access to health food options	Competence and confidence to deliver lifestyle messages, encourages behaviour change and signpost	Support and action to improve own lifestyle	Maximise benefit from existing resources e.g. advice on low or no-cost activity; walking to school; encourage increased use of existing community resources such as leisure centres Tackle health inequalities and wider determinants of health, through supporting individual changes e.g. help local populations manage debt, gain employment or tackle housing issues Equity of access, engaging those who will not have received support or considered accessing local services	Improves access to healthy lifestyle advice, improve morbidity and mortality risk factors within a local population, cost savings Provides approach, supports communities collaborating together

- **Screening tools:**
 - **Sensitivity** (how many detected) and **specificity** (how well people with condition correctly identified) important
- **VBA – Very brief advice:**
 - **Smoking:**
 - Advice to stop smoking has similar effect to no advice
 - 50% of long-term smokers die prematurely
 - Inhaled nicotine more addictive than heroin or cocaine
 - 70% want to stop smoking, 83% wish they'd never started, 30% try stopping each year
 - Cutting down smoking does not reduce risk proportionally e.g. going from 20 to 1 daily only decreases risk of CHD by 50%

- **TTFC – Time to first cigarette (upon waking):**

< 10/day	10-20/day	>20/day	
>2hrs	0.5-2hrs	<0.5hrs	Wakes at night to smoke
Low	Moderate	High	Very high addiction

- **Why do people keep smoking?**

- i) Habit, boredom, stress, social, taste, weight control
- ii) Nicotine binds to $\alpha 4\beta 2$ nicotinic receptor in the Ventral Tegmental Area (projecting to Nucleus Accumbens)
- **Problems with advice to stop:**
 - i) Negative message, nagging, nothing new, encourages conflict and denial, frustrating for both doctor and smoker, takes longer puts you off doing it again
- **Goals of behavioural support:**
 - i) Reduce motivation to smoke, bolster commitment to abstain, enhance ability to cope with cravings, ensure effective use of pharmacotherapy
- **Key elements of smoking VBA:**
 - i) Ask → Establish & record status, ex-smoker in last 3 years re-check status
 - ii) Advise → How? Combination of support and treatment
 - iii) Act → Mention specific service/person e.g. when ready book appointment with specialist nurse. If not ready, make clear door is always open.
- **VBA Benefits:**
 - i) 30 seconds
 - ii) 51x generates one long-term quitter
 - iii) Records smoking status (70% relapse rate)
 - iv) Positive, non-confrontational, opportunistic, informative, engaging
 - v) Evidence based
 - vi) Not a smoking cessation consult
- **Ideal smoker's pathway:**
 - i) Smoker attends GP for any problem → Smoking VBA → Quit attempt involving support and medication → Non-smoker
- **Quit statistics and epidemiology:**
 - i) 20-30% long term quit rate with support and treatment
 - ii) 3-5% long term quit rate with willpower
 - iii) Two or three quit attempts with support and treatment for 50% smokers to stop long term
 - iv) Smoking cessation with support and treatment is one of the most cost-effective interventions in healthcare
 - v) In isolation, behavioural support more effective than medication (but should use both)
- **With behavioural support:**
 - i) NRT – 23x more likely to quit
 - ii) Bupropion – 22x more likely to quit
 - iii) Varenicline – 11x more likely to quit
- **NHS Stop Smoking Services:**
 - i) Nurses, healthcare assistants, pharmacists or pharmacy assistants, specialist stop smoking advisers. Standard support regime 4-6 appointments over 6-12 weeks. Also NHS smokefree website and helpline.
- **Doctor-delivered smoking cessation:**
 - i) Tobacco dependence is a medical condition responding to medical treatment, doctors are independent prescribers used to tailoring treatment, cessation can be more effective from doctors than counsellors and nurses, smokers know trust and want help to stop from doctors.
- **Licensed pharmacotherapy for smoking cessation:**
 - i) Nicotine Replacement Therapy (NRT) – Patch and fast-acting gum, oral and nasal sprays, inhalers, lozenges and oral film. Every 2 weeks for 2-3 months, provides low level of nicotine, without the tar, carbon monoxide and poisonous chemicals in tobacco smoke; can help reduce unpleasant withdrawal effects; side effects: skin irritation, insomnia, upset stomach, dizziness, headaches
 - ii) Bupropion – Norepinephrine and dopamine reuptake inhibitor, non-nicotinic, used to treat depression; side effects: dry mouth, insomnia, headaches, constipation
 - iii) Varenicline (Champix) – Partial nicotine agonist ($\alpha 4\beta 2$ receptor), part stimulating to relieve cravings and withdrawal, part blocking to reduce pleasurable effects of smoking and risk of full relapse after temporary lapse, reduces cravings for nicotine and blocks the rewarding and reinforcing effects of smoking; side effects: feeling sick, insomnia, dry mouth, constipation, headache
- **Why should all three licensed pharmacotherapies be considered?**
 - i) What works well for one person may not be so good for another
 - ii) It allows the patient to choose the treatment they want (you guide, they decide)
 - iii) If the quit attempt fails, they know there are other options
- **E-cigarettes:**
 - i) 3 million UK users, 95% safer than smoking, at least as effective as NRT for smoking cessation, great potential for smoking cessation
 - ii) But long-term safety not known, half of users also smoke, controversial and many misconceptions
- **Don't challenge the addiction:**
 - i) VBA does not advise smokers to stop, ask how much or what they smoke or ask if they want to stop
- **Alcohol:**
 - IBA approach - Identification, brief advice, support
 - Units = (Volume x ABV)/1000 (N.B 568ml in pint)
 - **Alcoholism questionnaires:**
 - i) AUDIT-C, SADQ-C (Severity of Alcohol dependence questionnaire), CIWA-Ar (Clinical Institute Withdrawal Assessment for Alcohol-Revised)
 - ii) AUDIT-C score 1-4 sensible/ low risk, 5-7 hazardous/ increasing risk, 8-10 harmful/higher risk, 11-12 potentially addicted/dependent
 - **Implementation intention:**
 - i) What, where, when, and how e.g. I'm going to drink less beer – in pub, – whilst watching football, – by alternating alcoholic pint with non-alcoholic drink
 - **Self-help or referral:**
 - i) Patient information resources, apps, alcohol consumption trackers
 - ii) Referral to alcohol services if evidence of dependency. Automatic referral might lead to greater success.

The Coalition Government published their strategy on alcohol in 2012

ALCOHOL STRATEGY



There were broadly four areas of action...

1. National action on price, advertising and availability
2. Local action on licensing, enforcement and antisocial behaviour
3. Industry responsibility drive
4. Supporting for consumer information and individual behaviour change / treatment services



- **Physical activity and healthy eating:**
 - Physical inactivity may be as harmful as cigarette smoking and obesity
 - 1 in 4 women and 1 in 5 men were doing less than 30 mins a week of activity
 - 1 in 3 men and 2 in 5 women not active enough for good health
 - Inequalities by gender (women less likely to be active), socioeconomic status, disability, ethnicity (Mixed race individuals most active, Asian excluding Chinese least active)
- **Intervention:**
 - i) **Motivate, signpost, encourage, support. Ask, advise, assist.**
- **Increased relevance in:**
 - i) Hypertension, Type II diabetes, IHD, obesity, hypercholesterolemia
- **Screening tools:**
 - i) **Scot-PASW, GPPAQ** (General Practice Physical Activity Questionnaire), **MUST** (Malnutrition Universal Screening Tool)
- **Signposting/Support:**
 - i) Sports England, NHS, Couch to 5k, Parkrun, BBC sport
 - ii) Moving Medicine – Evidence based, condition-specific information about increasing physical activity based around 1 minute

Health coaching conversations

- 70% of global deaths due to cardiovascular disease, stroke, cancer, chronic disease and diabetes
- 80% of heart disease and stroke and 30-50% of cancers preventable (WHO)
- Risk factors: smoking, poor diet, lack of exercise and excessive alcohol consumption
- 2017 – Compendium on health coaching literature published, 2019 Addendum was published
- **Motivational Interviewing:**
 - Collaborative, person-centred form of guiding to elicit and strengthen motivation to change.
- **Person centred mindset/Spirit of Motivational Interviewing:**
 - **Compassion:** Desire to support patient's wellbeing
 - **Acceptance:** Respectful, non-judgmental approach, and values. Affirms the patient's autonomy and worth
 - **Partnership:** Working with patient, recognising the patient's resourceful expertise in their situation
 - **Evocation:** Helping patients identify their own perspectives and motivation for change.
- **Key health coaching questions:**
 - What's the issue?
 - What do you want ideally around this issue?
 - What part of this do you have control over?
 - What are your options going forward?
 - What is your next step?

Building rapport	Building rapport online
Be fully present Active listening Non-verbal cues e.g. particularly if writing notes	More time with introductions Slower clearer language and pauses Include arms and upper body on video for non-verbal cues Pay attention to non-verbal cues, prioritise looking at patient's face overlooking straight at camera Utilize technology e.g. whiteboard features for planning Uncluttered backgrounds & lights

- **Health coaching key aspects:**
 - Mutual trust and respect
 - Act as an enabler
 - Patients are resourceful experts
 - Solution-focused/ solution-oriented / future-focused
 - Non-judgmental
- **Coaching principles and issues:**
 - **Patient's assumed resourcefulness** – But think about capacity e.g. dementia or intoxication.
 - **Mutual trust and respect** – But not appropriate if you believe patient is dishonest, manipulative or they believe that of you.
 - Coaching is about **change and action** – Patient must at least partially want to address and explore the issue otherwise counter-productive. Remember always let patients know they can discuss same issue in future when they feel ready.
- **Clarifying expectations:**
 - **Practicalities:** location and timing of consultations
 - **Approach to confidentiality:** Mention from outset, and that may be broken in certain circumstances, e.g. risk of harm to patient or others.
 - **Expectations of the roles of practitioner and patient:**
Practitioner doesn't answer all questions or have simple 'fix',
 - shift from expert to enabler, patient sets own goals and solutions.
 - **Boundaries:** Clarify boundaries, your competence, and potential need to signpost
 - **Keeping a record:** Written record for you and patient to refer to and help keep sessions on track
- **OARS:**
 - **Open questions**
 - **Affirmations** – 'that's really good', 'you've been really brave to admit this'
 - **Reflective listening** – repeating patient's own language back to them or picking up on non-verbal cues e.g. 'I saw you smile'
 - when you mentioned that' or 'I noticed you said 'busy' a few times, do you mind letting me know some more about that?'
 - **Summarising**
- **Effective coaching questions:**
 - **Solution-orientated**
 - **Short and open**
 - Avoid multiple questions
 - 'What' rather than 'how' or 'why'
 - Pause
- **Four processes of motivational interviewing:**
 - **Engaging**
 - **Focusing**
 - **Evoking** – evoke thoughts and feelings about goal. Reinforce positive language – 'change talk'
 - **Planning** – lists, spider diagrams, self-monitoring
- **SMART goals:** Specific, Measurable, Achievable, Realistic, Timely

- **Effective goal setting:**
 - Important
 - Patient **motivated**
 - Aligns with **values** and situation
- Something patient works towards (not away from)
- Under **patient's control**
- **COM-B model:**
 - **Capability:** *Physical* skills and *psychological* knowledge
 - **Opportunity:** *Physical* restrains e.g. cost and time and *social* e.g. with companion or group
 - **Motivation:** *Reflective* considering long-term benefits and *automatic* seeing behaviour change as not only 'need' but also 'want'
- **Reviewing progress:**
 - **Acknowledge successes**
 - Facilitate **discussions** about what got in the way
 - **Support patient** to identify what they have learned and help them make further plans
- **Rolling with resistance:**
 - **Empathy**
 - Patient **autonomy**
 - Avoid arguing or advice giving



- Combined, this leads to behavioural change

- **Trust** and **collaborative** working
- **Strengthen belief** in **capacity** to change

Child health

Lifestyle	Behaviour	Social	Environmental
Foetal nutrition + maternal physical activity determines child weight Physical activity in children improved cardiovascular fitness/sleep/weight; learning/academic achievement and executive functioning; resilience, self-esteem, confidence, and trust in others; and reduced anxiety, stress, and depression Hidden hunger: over consumption of calories derived from cheaper, unhealthy, processed foods, which lack nutrients that are key to growth, brain development and immunity.	Inadequate foetal nutrition irreversible delayed cognitive/motor development If more physically active in childhood more likely as adult	Diet/physical activity/relationships/sleep/stress/socioeconomic status school readiness If child engages in inclusive + positive + meaningful physical activity develop physical literacy Dental caries= oral bacteria metabolise sugar and produce acid demineralising enamel and dentin from hard coating of tooth=> pain + infection + sepsis + death; Management may involve tooth extraction and prevention with sugar reduction and good oral hygiene Social deprivation associated with excessive intake of sugar; dental caries and obesity often occur together	Early interaction with parents/caregivers has heavy influence on lifestyle behaviours Parental obesity = biggest risk factor for childhood obesity

- **ACEs:**
 - Experiencing abuse: Physical, sexual, psychological or neglect
 - Witnessing domestic abuse
 - Having a close family member or caregiver who misused drugs or alcohol
 - Having a close family member or caregiver with mental health problems
 - Having a close family member or caregiver who served time in prison
 - Experiencing parental separation or divorce on account of relationship breakdown
- **Evidence behind impact**
 - Strong associations with **sexual risk taking, mental ill health, problematic alcohol use**
 - Strongest associations with **problematic drug use, violence**
 - Excessive **chronic activation of stress response** → **system dysregulation**
 - Childhood experiences affect **epigenome and gene expression** (temporarily/permanently)
- **Effective lifestyle medicine consultation:**
 - Be opportunistic
 - Be sensitive
 - Non-judgemental approach
 - Strengths-based approach (positive affirmations)
 - Focus on barriers that are particularly important to the child
 - Encourage children with long term conditions to be active (Moving Medicine)

RISK

Problematic drug use
Interpersonal and self-directed violence

Sexual risk taking
Mental ill health
Problematic alcohol use

Smoking, heavy alcohol use, poor self-rated health, cancer, heart disease, respiratory disease

Physical inactivity
Overweight or obesity
Diabetes

ACEs – but how?

Toxic Stress

Excessive chronic activation of the **stress response**

System dysregulation:
Child's developing brain, immune system, metabolic regulatory systems, and cardiovascular system

Epigenetics

Childhood experience (positive and negative) affects the 'epigenome' and gene expression.

Temporarily or permanently depending on how easily the gene is turned on or off.

Brain is developing most rapidly in early years so epigenetic expression shapes health and behaviour from this early stage.

We can reduce the impact of ACEs and intervention can prevent and counteract lasting harm. Early intervention is preferable.

Reducing stress, building responsive relationships and strengthening life and coping skills prevent the long-term effects of ACEs

- Relate to other children
- Engage parents and families
- Look at their strengths, needs, challenges and priorities
- Specific, short-term goals
- Daily routine – more effective goals
- **LEAP** criteria activities (Local, Enjoyable And Practical)

- **National child measurement programme:**
 - Measures height and weight in reception (ages 4 + 5) and year 6 (age 10 + 11)
 - Underweight: <2
 - Healthy weight: >2 and <85
 - Overweight: >85 and <95
 - Obese: >95
 - Severely obese: >99.6

	BMI centile criterion	Offer
Tier 1	none	• Universal prevention • Consultation with frontline health professionals • Public health approaches
Tier 2	≥ 91 st centile	• Intensive + community based • Time-specific • Public health approaches
Tier 3	≥ 99.6 th centile Or ≥ 91 st centile + co-morbidities/complex needs	• Specialist MDT • Weight management clinics • Longer term follow up
Tier 4	Not available in children	• Surgical options

Adolescent, maternal & women's health

- **Sexual health:**
 - Good sexual health = a state of physical, mental and social wellbeing in relation to sexuality
 - Requires a respectful approach to sexuality + sexual relationships, as well as the possibility of having pleasurable + safe sexual experiences free of coercion/discrimination/violence
 - Improves physical + mental health outcomes
 - Positively influences social + emotional wellbeing
- **Poor sexual health consequences:**
 - Unplanned pregnancies + abortions
 - Psychological consequences
 - STIs (chlamydia, gonorrhoea, genital warts and herpes)
- **Prevention:**
 - Contraception – discuss options + provide information + signpost
 - HPV immunisation → Cervical cancer/some mouth + throat cancers/some anal + genital cancers/genital warts
 - HIV prevention → PrEP/PEP
 - Education/health literacy
- **Lifestyle and sexual health:**
 - Sexual dysfunction
 - Smoking, prolonged alcohol misuse, stress, poor
 - Mental health
 - Risky sexual behaviour/sexual assault/sexual violence
 - Drug + alcohol use
- **Pre-conception interventions:**
 - **Maternal nutrition**
 - Folic acid before conception + up to 12 weeks pregnancy – reduces risk of neural tube defects
 - Green leafy veg, peas, kidney beans, chickpeas
 - Vitamin D throughout pregnancy + breastfeeding – promotes bone health
 - **Physical activity during pregnancy**
 - Decreases risk of hypertension in pregnancy, pre-eclampsia, gestational diabetes, severe depressive symptoms
 - Aim for at least 150 mins of moderate activity a week throughout pregnancy
 - **Alcohol during pregnancy**
 - Early miscarriage, premature birth, restricted growth, still birth, cognitive delay, foetal alcohol syndrome/spectrum disorders
 - **Smoking during pregnancy**
 - Miscarriage, premature birth, perinatal death
- **Foetal alcohol syndrome:**
 - Small stature
 - Facial features (microphthalmia, short palpebral fissures, epicanthal folds, a small or flat midface, a flat elongated philtrum, a thin upper lip, and small chin)
 - Abnormal palmar creases
 - Cardiac defects
 - Joint contractures
 - Cognitive deficits
- **Microbiome**
 - Majority of microbes found in the gut
 - Interacts with nervous + immune systems
 - **Factors influencing microbiome diversity + health**
 - Genes, age, environment, stress, diet, physical activity
 - Birthing process, Infant feeding method, Stress/diet/geography, Pharmaceuticals, Lifecycle changes, illness and disease, physical activity
 - **Produce secondary metabolites**
 - Detoxify toxins + breakdown fibre/resistant starches
 - Fermentation releases SCFAs
 - Source of energy/lowers colon pH/maintains immune homeostasis
 - Production of vitamin K + components of vitamin B
 - Gut mucosa tolerates beneficial micro-organisms while preventing pathogen overgrowth
 - Poor diet/unhealthy microbiome → low level, chronic inflammatory state/leaky gut (LPS's in bloodstream)
 - Can regulate innate + adaptive immune cells
 - Communication via ENS, vagus nerve and bloodstream
 - **Cytokines**
 - Two-way (both can induce stress mediated responses)
 - Involved in tryptophan metabolism → serotonin production)
 - **Early years**
 - Gut microbiome rapidly changing
 - **Adult years**
 - Gut microbiome relatively stable though affected by external factors
 - Chronic insult can cause harm
 - **Elderly years**
 - Microbiome becomes less diverse and more unstable with age

Adult & men's health

- **Self-care:**
 - "What people do for themselves to establish and maintain health and to prevent and deal with illness"
 - Can **modify** personal + intermediate **risk factors** for NCDs

- Help with **communicable disease** via **risk avoidance, good hygiene, health literacy**
- **7 pillars of self-care**
 - 1) **Knowledge and health literacy**
 - 2) **Mental wellbeing**
 - 3) **Physical activity**
 - 4) **Healthy eating**
 - 5) **Risk avoidance**
 - 6) **Good hygiene**
 - 7) **Rational use of products and services**
- **Self-care continuum**
 - Healthy behaviours → acute conditions + trauma
- **Gender vs sex:**
 - **Sex** - biological aspects demonstrated by their anatomy, produced by their chromosomes, hormones, and their interactions
 - **Gender** - social construction relating to behaviours and attributes based on labels of masculinity and femininity
- **Drug misuse:**
 - **Risk factors**
 - **ACEs**
 - Struggling with **mental health**
 - Apart from these, very diverse
 - **Impacts**
 - **Smoked** – **lung issues**
 - **Injected** – **blood clots, bacterial skin infections, viral infections** (HIV, Hep C)
 - Struggle with formal work
- **Alcohol misuse:** (*>14 units per week or binge drinking (>8 units for men or >6 for women)*)
 - **Risk factors**
 - **Men + middle aged**
 - **Less deprived** (although more harm to more deprived, due to **multiplier effect** – **alcohol harm paradox**)
 - **Impacts**
 - Relationship breakdown, domestic abuse, crime, road traffic accidents
 - **Support + treatments**
 - Brief advice + self-help + follow up (If unsuccessful – extended brief intervention/CBT)
 - Planned withdrawal (primary care/specialist)
 - Screening tools = AUDIT and **AUDIT C**
- **Acute withdrawal symptoms** – admit to hospital!
 - **Hypertension, tachycardia, sweating, tremor, agitation**
 - **Delirium tremens: confusion, seizures, hallucinations, hyperthermia**
 - **Risk factor for**
 - **Wernicke's Encephalopathy** (**thiamine/vitamin B1 deficiency**)
 - Neurological emergency!
 - Symptoms: gait and balance disturbance, altered consciousness, eye movement abnormalities
- **Relationship between alcohol and mental health**
 - Maladaptive coping strategy → increased alcohol consumption depressant effects of alcohol → poor mental health → maladaptive coping strategy (cycle)
- **Reward pathway:**
 - Behaviours of addiction hijack our reward system to produce greater **dopamine** in the **nucleus accumbens**. This increases activation of dopamine receptors, particularly D1 receptors and also has an impact on memory and conditioning via pathways involving the amygdala, orbitofrontal cortex, and hippocampus
 - Drivers of addiction:
 - Reward process; withdrawal; cravings
- **Withdrawal:**
 - Driven by recruitment of stress areas of the brain and increase in **stress related neurochemicals** - **corticotrophin releasing factor** (CRF), **norepinephrine**, **dynorphin** in **extended amygdala**. There is also **decrease in activation of reward systems**- creating **negative reinforcement** whereby the user seeks the drug to prevent the effects of withdrawal, to achieve emotional equilibrium, rather than a euphoric high.
- **Stress:**
 - **Mechanisms of stress:**
 - Oxidative stress
 - Immune dysregulation
 - Inflammation
 - Chronic disease
 - **Sympathetic nervous system in 'fight or flight'**
 - Production of cortisol takes priority over other steroids / hormones
 - Hormone production interconnected
 - Can also lead to hormone dysregulation
- **FITT**
 - FITT principle (exercise prescription to help participants understand how long and hard they should exercise for)
 - Frequency
 - Intensity
- Daily choices → lifestyle → self-managed ailments → minor ailments → long term conditions → acute conditions → compulsory psychiatric care → major trauma
- **Behaviours**
 - Activation, Incentivisation, Nudges, Gamification
- **Personalisation**
 - E.g. lifestyle medicine, HiAP approach
- **Technologies**
 - Health apps/wearables
- Less able to look after others
- **Violence and abuse** associated with drug trade
- **Support + treatments**
 - GP, secondary care, psychiatry services,
- **Specialist services**
 - **Methadone** - **heroin substitute**
 - Non-medication pain management
- **Mental illness**
- **Abuse during childhood**
- Reducing doses of **chlordiazepoxide** (protect against withdrawal symptoms) then **acamprosate**
- **Maintaining alcohol abstinence** –acamprosate reduces urge to drink and works best in tandem with psychological therapy, influences **GABA**
- Likely require IV thiamine
- Prior to start planned withdrawal → inform DVLA and stop driving & assess for alcohol related comorbidities (pancreatitis, MH, liver disease)

- Time
- Type
- **Stress** can contribute to **chronic disease** through key biological processes -can be mitigated by addressing factors e.g. physical activity, diet, sleep, and social connection

Older people's health

- **Malnutrition:** diet does not contain the right amount of nutrients, which may be in the form of under-nutrition or over-nutrition
 - Not a normal part of ageing
 - **Ageing**
 - decreased food intake; decreased basal metabolic rate, lean body mass, sense of taste/smell
 - **Causes:**
 - Poor dentition
 - Difficulty swallowing
 - Declining coordination + eyesight
 - Arthritis
 - Low mood
 - Cognitive decline
 - Cancer, heart failure, CKD, hypothyroidism
 - Coeliac disease + IBD => malabsorption
 - Medication (diuretics, antihypertensives)
 - Reduced physical activity
 - **Effects:**
 - Impaired immune function
 - Poor wound healing
 - Osteoporosis
 - Cognitive impairment
 - Mood disturbance
 - Joint + muscle pain
 - Calcium, vitamin D/C/B12, folate deficiency
 - Osteomalacia (vit D + Ca2+), anaemia (vit B12 + folate), sarcopenia, poor wound healing (vit C)
 - Overnutrition in older people (physical or cognitive problems lead to decreased activity and weight gain)
 - **Assessment:**
 - Changed weight/BMI
 - Mental health
 - Social/drug/past medical history
 - Food and fluid diary
 - MUST= malnutrition universal screening tool
 - **Support:**
 - Doctors, dieticians, OT, social care, speech and language therapist, dentists
- **OSA**
 - Results from a **partial/fully blocked airway**, Tonsils too large, jaw set too far back, excess weight
 - If there's a lot of adipose tissue at base of tongue and around pharynx then it's more likely airway will collapse when muscles are relaxed i.e. during sleep
 - **Airway blocked in sleep → brain hypoxia → mini arousal → sharp intake of breath**
 - **↑ risk of CVD/stroke/type 2 diabetes/severe depression/severe COVID-19**
 - **Presentation + diagnosis**
 - **Loud snoring, pause in breathing, large gasp of air**
 - Pauses in breathing lead to **hypercapnia**, increased CO2 in brain and so cellular death
 - **Excessive daytime sleepiness** (EDS)!
 - Ideally diagnosed with PSD – expensive + resource intensive
 - Always consider bed partner!
 - Use **STOP-Bang questionnaire** (snoring, tired, observed, pressure, BMI, age, neck size, gender)
 - **Risk factors**
 - Sleeping on back
 - Anything impacting REM sleep (muscles are paralysed)
 - Alcohol, some antidepressants
 - splint in airway to keep it open
 - **Loss weight, quit smoking, reduce alcohol**
 - **Treatment**
 - **CPAP** – gold standard but low compliance; pneumatic
- **Insomnia** (Predictor of depression – be wary)
 - **Causes**
 - Continuously heightened **stress** → **hyperarousal therapy**
 - **Diagnosis**
 - Difficulty initiating/maintaining sleep
 - Waking up earlier + resistance going to bed
 - Difficulty sleeping without parent/caregiver
 - Daytime difficulties due to sleep
 - 1-3 months = short term, 3 months = chronic
 - **Treatment**
 - Psychotropic medication not recommended – side effects can be worse
 - First line = CBT
 - Sleep hygiene, relaxation training, paradoxical intention, sleep restriction therapy
- **Sleep difficulties** = Stress, mental health problems, physical inactivity, diet, chronic conditions, drugs, alcohol, physiology (melatonin production decreases, sleep more fragmented),
- **Hypnotics**
 - **Benzodiazepines** = GABA receptor agonists
 - have anxiolytic, hypnotic and sedative effects
 - **short term** management of insomnia, avoid dependence, tolerance, SE (hangover effect- drowsiness, unsteady gait, slurred speech, confusion)
 - Overdose in elderly more susceptible to cardiorespiratory depression
 - Zopiclone is an example of 'Z drug' à a **non-benzodiazepine hypnotic**
- **Social isolation and loneliness**
 - **Social isolation:** objective measure of number of contacts that a person has; quantity
 - **Loneliness:** subjective feeling about the gap between a person's desired levels of social contact compared to their actual level of social contact; quality; measurement: community life survey, UCLA 3
 - **Health impacts:**
 - **Cardiovascular:** hypertension, coronary heart disease, stroke
 - **Immune system:** increased inflammatory response to stress, more susceptible to infections
 - **Neuroendocrine:** type 2 diabetes, increased cortisol levels due to dysregulated stress response • Cognition: impaired sleep quality, memory problems, increased risk of Alzheimer's disease

- **Psychological:** depression, anxiety, increased risk of suicide
- **Social prescribing** = Enables GPs, nurses + other primary care professionals to refer people to local, non-clinical services
 - **GP** assesses patient needs and refers to **link worker**, who will signpost to a **community organisation** and follow-up the patient
 - Improves QoL + mental health
 - Reduction in use of NHS services
 - Hard to measure outcomes due to the heterogenous nature of the programmes/service users/outcomes
 - Without strong evidence to support, hard to commission social prescribing programmes
 - Risk of seeming like you are trivialising the patient's issues
 - A change in NHS culture, strong community links, a holistic approach to patient care, proper investment, and accurate measures of success
- **Green spaces**
 - **Physiological changes** – Reduced HR, Changes in HRV, Reduced salivary cortisol and Reduced diastolic blood pressure.
 - **Physical health** – Decreased risk of T2DM, Reduced incidence of stroke, hypertension, dyslipidaemia, asthma and CHD, Decreased risk of pre-term birth and small size gestational age and Decreased risk of cardiovascular + all-cause mortality.
 - **Mental health** – Reduced feelings of stress, Reduced anxiety and depression, Improved attention and memory and Improved childhood development.
 - **Social** – Increased job opportunities, Improved work productivity, Reduced crimes and Improved social cohesion + interaction.
- **Physical activity**
 - **Anti-inflammatory** effects
 - Reduced inflammatory visceral fat
 - Anti-inflammatory **myokines** during muscle contraction
 - Effects at **mitochondrial** level
 - Increased charge the mitochondria during sedentary behaviour increases 'electrochemical charge' across the mitochondrial membrane → leakage of free radicals into the cell → cell damage → inflammation → telomere shortening → ageing.
 - ATP use during muscle contraction **reduces charge in mitochondria** → **reverses inflammation + telomere shortening** (slows ageing process)
 - Increases no. + size of mitochondria
 - Weight loss
 - Benefits of being active + overweight outweigh benefits of weight loss while being inactive
 - Coronary artery disease - plaque regression
 - Reduces blood pressure
 - Can prevent rehospitalisation after an MI
 - Diabetes
 - Aerobic/resistance exercise reduces mortality and increases insulin sensitivity
 - Better HbA1c/blood lipids/waist circumference/BMI
 - Effect on cells lining blood vessels, allowing **vasodilation**, reducing inflammation
 - Effect on overall **anti-inflammation** and **antioxidant** systems
 - Screening tool = **Scot-PASQ and GPPAQ**

Global health

Key Words

- Global South Countries
 - Less Economically Developed Country
 - Emerging Market
 - WASH = Water, Sanitation and Hygiene
 - Communicable diseases
 - Non – communicable disease
 - Polio Global Eradication Initiative
 - WHO
 - International Health Regulations
 - Human Development Index (HDI)
 - HIC (high income country)
 - LMIC (low- and middle-income country)
 - Sustainable Development Goals
 - Maternal death
 - MMR = Maternal Mortality Ratio
 - Planetary Health
 - One Health
 - Ecosystem
 - Ecosystem Services
 - Biodiversity
 - Wetland
 - Representative Concentration Pathway (RCP)
 - Social-Ecological Systems
 - REDD+
 - Externalities
 - Circular economy
 - State shift
 - Resilience
 - Human Health, Animal Health, Plant Health
 - Environmental Sustainability
 - AMR (Antimicrobial Resistance)
- *an area of study, research and practice that places a priority on improving health and achieving health equity for all people worldwide*

HDI & how it relates to different Countries:

- HDI = composite continuous value between 0.001 and 0.999 that's used by UN to quantify development and it's based off **three measures of development**:
 - Knowledge:** Based on education and literacy
 - Healthy life:** Based around life expectancy
 - Standard of living:** Based on income adjusted for local circumstances (purchasing power parity, PPP)
- Different Levels of HDI:
 - Very High: Norway, Ireland, and Switzerland
 - High: Seychelles, Trinidad and Tobago, and Albania
 - Medium: Kyrgyzstan, Morocco, and Guyana
 - Low: Mauritania, Benin, and Uganda

Global South Countries*	Third World	Less Economically Developed Countries	Emerging Market
countries where income tends to be low + economies are recently industrialising Many/most countries often have a colonial history Geographically most Global South countries are around the tropics (not necessarily in the southern hemisphere)	In the era of the Cold War, countries that were aligned with NATO were deemed first world. Countries aligned with the USSR were considered second world. This term was originally intended to denote non-aligned national state actors.	A term that has been vaguely used to categorise countries on then basis of their gross national income (GNI). This term has now been removed from routine use by the World Bank.	An economic term suggesting that a country is about to engage with global markets. This term remains commonly used in business with pharmaceutical companies often using it to describe multiple countries where business opportunities are anticipated.

* Australia and New Zealand are not considered Global South countries

- **WASH = Water Sanitation and Hygiene**

- Water = safe water for drinking washing and domestic activities
- Sanitation = safe removal of waste including toilets waste disposal and
- Hygiene = a range of so-called health promotion activities that encourage protective healthy behaviour practices
- Issues related to WASH:

- Water-related infectious diseases:

- Water-borne (drink contaminated water/eat food prepared with contaminated water)
 - e.g. Cholera & Diarrhoeal Disease
- Water-washed (poor personal hygiene)
 - e.g. Scabies & Trachoma
- Water-based (transported by aquatic organisms)
 - e.g. schistosomiasis & guinea worm
- Water-related – spread by insect vectors
 - e.g. malaria & river blindness
- Diseases caused by poor sanitation
 - e.g. hookworm

- 2.1bn people (30% of world's population) lack access to safely manage water
- 4.5bn people (60% of world's population) lack access to effective sanitation

- Globally Prevalent WASH-related diseases:

- Diarrhoea, Malaria, Schistosomiasis, Trachoma, Intestinal Helminths, Japanese Encephalitis, Hepatitis A



- **Maternal Death**

- death of a woman “while pregnant or within 42 days of termination of pregnancy... from any cause by the pregnancy or its management but not from accidental or incidental causes.”
- Maternal mortality ratio is the epidemiological indicator commonly used
- Maternal mortality ratio = (Number of maternal deaths / Number of live births*) X 100,000**
 - *number of live births (in the same period)

- **SDGs (Sustainable Development Goals)**

- 17 Goals set in 2015 designed to be achieved by 2030 (within 15 years) to transform human progress at a global level
- The Goals are universal & relevant for High-income countries & LMICs

- **REDD+ (Reducing Emissions from Deforestation and Forest Degradation (REDD))**

- tries to assign a financial value to the carbon stored in trees to help developing countries invest in low-carbon paths to sustainable development
- REDD+ includes an added focus on conservation, sustainable management of forests, and enhancement of forest carbon stocks

- **Representative Concentration Pathway (RCP):** 4 trajectories of the concentrations of greenhouse gases in the atmosphere consistent with a range of possible future emissions - identified by their approximate total radiative forcing (i.e., warming effect) in the year 2100

- **Social-ecological systems:** Natural systems do not exist without people and social systems cannot exist totally in isolation from nature. These systems are truly interconnected and coevolve across spatial and temporal scales.



Externalities	State shift	Resilience	Circular economy
A benefit or cost that affects an individual or group of people who did not choose to incur that benefit or cost.	Large, lasting changes in the structure and function of social-ecological systems, with substantial impacts on the ecosystem services provided by these systems.	“the capacity of any entity—an individual, a community, an organization, or a natural system—to prepare for disruptions, to recover from shocks and stresses, and to adapt and grow from a disruptive experience.”	A global economic model that decouples economic growth and development from the consumption of finite resources. Circular economy systems keep products in use for as long as possible, allow for the recycling of end products, and eliminate waste.

- **Ecosystem:** Dynamic complex of plant, animal, and microorganism communities and the non-living environment acting as a functional unit
- **Ecosystem services:** Benefits provided by ecosystems to humans e.g. products – food water, regulation of floods, soils erosions and disease outbreaks & non-material e.g. recreational & spiritual benefits of natural areas

Holocene	Anthropocene
A geological epoch that began about 11 700 years ago and encompasses most of the time period during which humanity has grown and developed, including all its written history and development of major civilisations	Proposed name for a new geological epoch demarcated as the time when human activities began to have a substantial global effect on the Earth's systems. The Anthropocene is not yet formally recognised as a new geological epoch and several dates have been put forward to mark its beginning.

- **Biodiversity:** biodiversity means the variability among living organisms from all sources, This variability includes diversity within species, between species, and of ecosystems.

Planetary Health and Sustainability:

Conceptual and empathy failures (imagination challenges)	Knowledge failures (research and information challenges)	Implementation failures (governance challenges)
Over-reliance on gross domestic product as a measure of human progress Failure to account for future health and environmental harms over present day gains Disproportionate effect of those harms on the poor and those in developing nations	Failure to address social and environmental drivers of ill health Historical scarcity of transdisciplinary research and funding Unwillingness or inability to deal with uncertainty within decision making frameworks	Governments and institutions delay recognition and responses to threats especially when faced with uncertainties, pooled common resources, and time lags between action and effect.

Power & politics

Key words

- Hedonic Calculus
- Liberal
- Conservative
- Libertarian
- Collectivist
- The Political Compass
- Beauchamp and Childress' Taxonomy of Medical Ethics
- Autonomy
- Non-Maleficence
- Beneficence
- Justice
- Consequentialism
- Utilitarianism
- Distributive Justice
- Social justice
- Minimum Unit Pricing (MUP)
- Responsibility Deal
- Stakeholder
- Partner
- Veil of Ignorance
- Power
- Authority
- Coalitions
- Incremental change
- Transformative change
- Tame Problem
- Wicked Problem
- Plan
- Strategy
- Third Sector Groups
- Groupthink
- Politics
- politics
- Harm Principle
- Objectives (tactical)
- Aim (strategic)
- Collaboration (Coalitions)
- Compromise
- Advocacy
- Tudor-Hart's inverse care law
- WHO CODE OF PRACTICE (2010)**
- UK CODE OF PRACTICE (2021)**

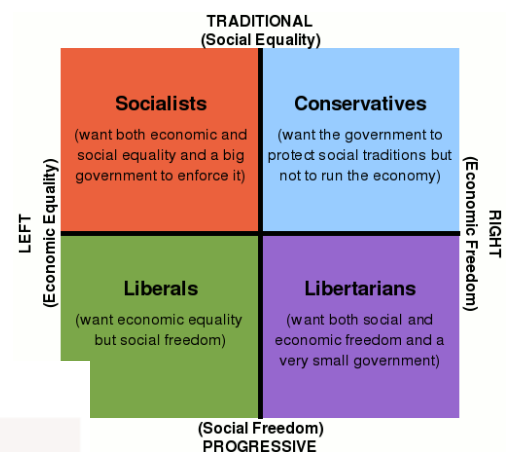
Politics, politics, and Political Ideologies:

- Politics**
 - political parties and the philosophy of government, driven by politicians – typically-elected, decision-makers who take countries forward
- politics**
 - process of us gaining or applying power in our local group or organisation

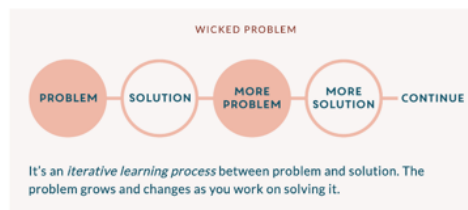
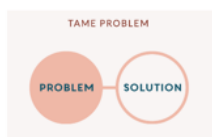
Stakeholders, Partners & Coalitions:

- Stakeholder**
 - individual or organisation that has an interest in what you're doing
- Partner**
 - a specific subgroup of stakeholders who are central to the success of whatever it is you're trying to achieve

Coalition = a group of stakeholders who work together for a shared goal
- Power**
 - the ability to exercise one's will over others



TYPES OF PROBLEM



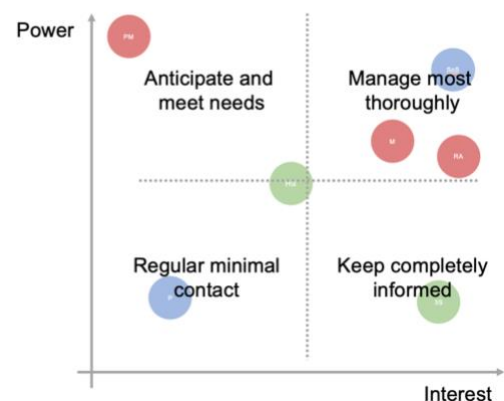
Tame problem	Wicked problem
Can be simple or complex, but is prone to being solved by a process of linear solution.	Is complex and complicated by multiple competing drivers, interests, uncertainty and possible competing solutions
We are typically trained to take a problem-solving approach that will, for the most part, work for these...	Many public policy issues and social protection issues (including health care) are examples of this.
They require a plan ...	They require a strategy ...

- Types of Change:
 - Incremental** (step by step – typically slow)
 - Transformative** (root and branch change, usually fast)

Stakeholder Analysis:

- High Power + Low Interest e.g. PM
 - i) Anticipate & Meet Needs
- High Power & High Interest e.g. Secretary of State, Media, Retailer's Association
 - i) Manage Most Thoroughly
- Low Power & Low Interest e.g. the Public
 - i) regular minimal contact
- Low Power & High Interest – e.g. Third Sector Groups
 - i) Keep Completely Informed

STAKEHOLDER MAP



Guiding principles

There are five guiding principles that underpin the Code of Practice.

1. International migration of health and social care personnel can make a contribution to the development and strengthening of health and social care systems to both countries of origin and destination countries if recruitment is managed properly.
2. Opportunities exist for individuals, organisations and health and care systems to train and educate and enhance their clinical practice.
3. There must be no active international recruitment from countries on the list, unless there is an explicit government-to-government agreement with the UK to support managed recruitment activities that are undertaken strictly in compliance with the terms of that agreement.
4. Recruitment of international health and social care personnel is closely monitored and reported on to the Cross Whitehall International Recruitment Steering Group.
5. International health and social care personnel will have the same legal rights and responsibilities as domestically trained staff in all terms of employment and conditions of work. They will also have the same access to further education and training and continuous professional development.

WHO CODE OF PRACTICE (2010)

The Code aims to establish and promote **voluntary principles and practices for the ethical international recruitment of health personnel and to facilitate the strengthening of health systems.**

Member States should discourage active recruitment of health personnel from developing countries facing critical shortages of health workers.

The Code was designed by Member States to serve as a continuous and dynamic framework for global dialogue and cooperation.

Figure 2 UK CODE OF PRACTICE (2021)

Nuffield Bioethics Ladder	Examples
8. Eliminate choice	Make the consumption of novel psychoactive substances criminal
7. Restrict choice	Restrict the sale of electronic cigarettes to those over the age of 21
6. Guide choice through disincentives	Increase the cost of tobacco products
5. Guide choices through incentives	Subsidise gym memberships for people earning under £30,000 per year
4. Guide choices by changing the default	Auto-enrol health care staff into the NHS pension unless they opt out
3. Enable choice	Publish data on mortality and patient experience for all NHS hospitals and allow patients the freedom to select their own hospital provider
2. Provide information	Introduce traffic-light food labelling
1. Do nothing or monitor	-

Beauchamp & Childress's taxonomy of medical ethics	Utilitarianism	Veil of Ignorance	Harm Principle
1. autonomy (freedom to act independently) 2. non-maleficence (do no harm) 3. beneficence (do good) 4. justice (fairness)	Most Happiness & Least Unhappiness for the Largest Number of People	If you had no idea what class, position, or social status you were going to be born into - what type of society/social contract would you want? Would you want a society with unlimited opportunity for a privileged few? Or a society with equality of opportunity for all.	"The only purpose for which power can be rightfully exercised over any member of a civilized community, against his will, is to prevent harm to others."